

Green Production Building—Moving Ducts Inside

Savvy builders can finance green features with the money homeowners would have spent on higher utility bills.

by **Ryan Kerr**

The green production builder is responding to clients' wishes to build with the new goals of promoting better occupant health and environmental stewardship. The best part? Savvy builders can finance green features with the money homeowners would have spent on higher utility bills. The utility bill savings are generated by what is possibly the greenest building practice of all—energy efficiency.

The Building Industry Research Alliance (BIRA) is focused on supporting the design and construction of near zero energy homes as part of DOE's Building America program. As a member of BIRA, I have the privilege of working with some of the brightest building industry professionals to study advanced technologies and designs that promote Building America's ideal of net zero energy homes.

Even when energy-saving features do in fact save energy, additional work is needed to make them cost-effective in production home construction. The first step is to move from design to practice. The second step is market transformation. And market transformation is taking hold in Washington State, as two production builders implement a highly efficient, yet underutilized, design concept: moving ducts inside conditioned space (see "Chasing Interior Ducts," *HE* May/June '02, p. 24).

On a trip to visit BIRA research partners in the Pacific Northwest, I met with two production home builders who were moving all their homes'

HVAC systems out of unconditioned attics, crawlspaces, and garages and into interior spaces. From the theoretical perspectives of technology and design, this technique is neither new nor advanced. From the construction perspective, however, it is hardly a mainstream practice, and it represents a very significant step toward increasing a home's heating and cooling energy efficiency.

In the world of production building, change means time, liability, and money. Thus builders are reluctant to adopt any new technique—but now that home buyers are beginning to recognize the value of energy efficiency, change can mean profit. The techniques used by our two builder partners are brilliant in their simplicity. And they are affordable, both in first costs, because they make use of construction efficiencies, and in operational costs, because they save energy.

Quadrant Homes

Quadrant Homes began building homes in the Seattle area in the 1960s. Today it builds more than 1,000 homes a year, mainly for entry level buyers. Quadrant is the largest builder in its geographic region and boasts industry-leading annual profit margins.

Quadrant Homes has obtained notable market results using an even-flow predictable scheduling scheme, in which it begins seven homes per day and finishes each house in precisely 54 days.

Using a standardized approach, Quadrant has achieved impressive results in both envelope and duct tightness testing, as performed by Washington State University (WSU) in both Energy Star and non-Energy Star homes. Quadrant's quality control is generally excellent. Although it can be challenging for a production builder to change methods, once those methods are changed, the improvements can be standardized and implemented across all homes, thereby improving thousands of homes.

One of Quadrant's subcontractors praises the even-flow production protocols, saying, "You know the job will be ready, you know exactly how much time you have, and you know you'll be alone to work in the absence of other subs." Another aspect in Quadrant's favor is its flexibility. Being open to innovation in the constantly changing world of green building gives the Quadrant homes an edge on the future.

What gave Quadrant the idea of moving its ducts inside? A research team from WSU was responsible for this suggestion. The team, consisting of Andrew Gordon, Chuck Murray, Mike Lubliner, and David Hales, approached Quadrant with the idea of moving ducts inside. In the broader context, Andrew Gordon was investigating how a prescriptive approach could be taken to reach 50% energy savings in heating and cooling and thus qualify Quadrant for a \$2,000 federal tax credit for each home. As Andrew discovered,



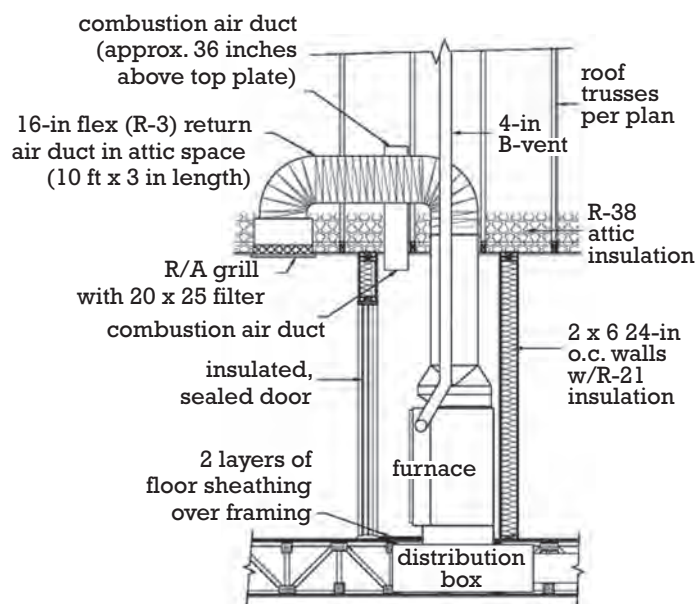
The ducts in this Vancouver, Canada, home are between floors and hidden in chases.

NEW TRADITION HOMES



These New Tradition homes are two-story, with ductwork located between floors.

Quadrant Homes Typical Upper Floor Furnace Room Section



QUADRANT HOMES

Figure 1. Supply ducts are all located between the first and second floors, running parallel and perpendicular to the open-web, or parallel flat chord trusses.

building energy simulation software tools continually pointed to decreasing envelope leakage and moving ducts inside as the best options for cost-effective energy savings that were above Energy Star requirements.

Quadrant already had the tight envelope, and it happened to be in the midst of redrafting its home designs. As Quadrant was testing the waters with tax credit homes—Energy Star homes with added attic insulation and an

improved AFUE (annual fuel utilization efficiency) furnace—it was the right time to make wholesale changes to the company's HVAC designs. The next time Andrew met with Quadrant's design team, they had worked with their HVAC subcontractor, Bob's Heating, to design a new system (see Figure 1).

The most important element of the new design is that the supply ducts are all located between the first

and second floors, running parallel and perpendicular to the open-web, or parallel flat chord trusses. The open-web trusses allow Quadrant subs the flexibility to run plumbing, electrical, and ductwork through the open webbing between the second-floor trusses.

Quadrant switched to open-web trusses five years ago in order to panelize the second floor in its homes, thereby saving a day per home in its tight construction schedule. The additional costs for the new truss design were offset by cost savings in the HVAC system equipment and installation. These cost savings were made possible by shorter uninsulated duct runs and the purchase of smaller, less expensive furnaces. These in turn were made possible by the reduced heating load caused by moving the HVAC system inside.

Although Quadrant no longer panelizes its floors, it has stuck with the open-web trusses because the subs wouldn't have it any other way. The 18-inch open-web trusses also streamline construction when buyers choose optional floor plans that require longer second-floor truss spans. The longer truss spans do not require additional bearing walls, as they did when the design called for I-joint trusses.

Quadrant also moved its furnace from the attic to a mechanical closet on the second floor. According to WSU's David Hales, it is better for the homeowner to have the air handler located in a mechanical room, where service or replacement is easier. The mechanical closet is actually built to outside-space code requirements (with insulated walls and door) to accommodate an 80-AFUE furnace with nondirect venting. A 90+ AFUE furnace, as required by Energy Star, would not require this feature, although it would require other features, such as a gas condensate drain. Quadrant is not prepared to make a wholesale upgrade to 90-AFUE furnaces, mainly because an improved furnace is one of many optional energy-efficient upgrades that it offers to its buyers.

HVAC

Quadrant's return ductwork is still in the attic. The duct is insulated to R-8, per Washington Building Code, and surrounded by additional insulation in some areas. This configuration is difficult to model using many building energy simulation tools, but with all the supply ducts and the furnace within conditioned space and minimal return duct in the attic, this HVAC system is greatly improved.

The HVAC contractor. One of the most important elements in making a smooth transition to a new building practice is close coordination among the affected trades. Wade Craig, of Bob's Heating, began working with Quadrant eight years ago to introduce 90-AFUE furnaces as optional equipment. As customers began to choose this option, Quadrant looked to Energy Star for guidance and today, as we have seen, it is moving its ducts inside.

When I asked Wade about moving ducts inside, his first response was "Anything we can do to get our guys out of the crawlspace is great." According to Wade, ten years ago almost all homes in Washington had ducts in the crawlspace. This is a problem for several reasons, not the least of which is the working conditions, and subsequent quality issues, for the HVAC trades. It is easier for installers to do a quality job in the home when they are not lying in mud, deep in a corner of the crawlspace, where they know an inspector will not crawl around to inspect their sealing around boots and the like.

In addition, when ducts are installed between floors they are clearly visible to the general contractor and inspector. Another major concern is trade damage or trade fighting, which occurs when electrical or plumbing subs compromise ducts to squeeze their work into place. With open-web trusses there is enough room for all trades, making each sub's work easier, and of higher quality.

Wade says his firm can make more economical bids for HVAC work when ducts are inside, as they save on material (the runs are shorter), save on insulation (there's no need for

insulation inside), and save on labor (it's easier).

I asked Wade what he would tell builders who were considering moving to open-web trusses and moving the HVAC system inside. Here are his two biggest pieces of advice:

1. Don't be afraid of the additional costs for open-web trusses. This includes the cost of adding at least 4 inches to the height of the home, to make room for ducts. The additional costs will pay for themselves for many reasons. Callbacks will decrease; subs can coordinate their work more easily; many trades can bid less for jobs because it is easier to run equipment between floors; and builders can save on materials, because open-web trusses allow for longer floor truss spans, as explained above.
2. Work with your HVAC sub to design a system that works for both of you. This is imperative for any design change that influences the way subs complete their work.

Caveats. Many builders assume that with ducts inside, duct leakage is less important. However, problems can occur when the ducts are located between floors and the rim joists are not properly insulated and sealed. Ducts must be well-sealed with mastic, and tested with a Duct Blaster for quality assurance. Duct leakage could result in significant losses to the outside through thermal and air leaks. If the ducts leak, rooms farthest from the air handler may not receive adequate air flow. There are excellent sealing and insulating protocols in the manuals referenced at the end of this article.



The duct in this New Tradition home under construction is a trunk between floors that supplies the first floor through ceiling registers and the second floor through floor registers.

Factors in Quadrant's easy transition. Ready for innovation and open to change, Quadrant is flexible, and its flexibility has allowed it to evolve. These factors also had a hand in Quadrant's easy transition:

- Quadrant's HVAC sub is a champion for moving ducts inside and was involved in original design review.
- Quadrant builds entry-level homes with few architectural elements that could complicate sealing and HVAC design.
- Quadrant was involved in a redraft of designs when WSU suggested moving the ducts inside.
- Quadrant was already using open-web floor trusses, which made the transition easier.
- Quadrant's systematic approach to building made it easier to effect a wholesale change.

New Tradition Homes

New Tradition Homes, headquartered in Vancouver, Washington, serves new home buyers in the western and central parts of the state. New Tradition Homes has been a Building America partner for several years and is one of the few builders we work with that have an

Thermal Barrier and Duct Locations

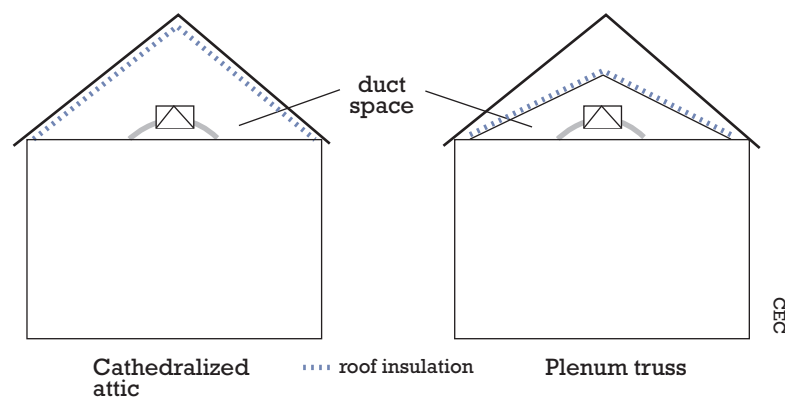


Figure 2. Here is a schematic of the thermal barrier and duct locations for keeping ducts inside conditioned spaces while using a cathedralized attic or a plenum truss.

integral building science team whose members represent a cross section of the company's operations. This structure enables the builder to make smooth transitions to cost-effective design changes. New Tradition Homes works with Building Knowledge, a residential building science firm, as well as WSU and BIRA, to identify, test, and implement new designs and technologies. New Tradition built over 400 homes in 2005, but it has decreased its construction pace in recent years with the tightening of the local home market.

Like Quadrant, New Tradition has begun to move ducts from the crawlspace and attic to the inside of its homes. It began by planning the move with its HVAC subcontractor. With planning, the process was relatively painless for New Tradition, with cost increases estimated at about \$500 per home. However, with future redesigns, New Tradition Homes hopes to eliminate cost increases or actually to save money by moving the ducts inside. The loss of floor area was not an issue for New Tradition, as the homes are two-story models, with the ductwork located between floors. The air handler was included in the envelope by extending the conditioned space into the garage. Indeed, the only difficulty was that New Tradition wanted to integrate the improvements into homes already under construction and was unable to do so. This illustrates the need to address HVAC improvements

in the design, rather than in the building, phase.

New Tradition Homes runs a supply duct perpendicular to and beneath the first-floor trusses. New Tradition uses a typical closed-web truss that makes running large ducts perpendicular to trusses within the floor cavity nearly impossible (see photo, p. 26). Given that the duct is underneath the first-floor trusses, an air barrier is not required as it would be if the duct were underneath the second-floor ceiling. However, the first-floor trunk required additional coordination between the HVAC sub and the framing sub, because the ducts went in after framing was complete and the framer then returned to frame the ceiling chase around the duct. This process is easier for a production builder than it is for a custom builder, because in production building, the trades are typically working on other homes in the same community and can easily coordinate to come back and finish the job.

Caveats. The same caveats apply to New Tradition as to Quadrant. Proper sealing of ducts and rim joists is essential. Here again, ducts should be tested for leakage (before the drywall goes in, while remedial actions can be taken), and the envelope should be tested for airtightness.

Factors in New Tradition's easy transition. Again, a willingness to try something new was just as important as having the tools and skills with

which to do the work. These factors also contributed to New Tradition's easy transition:

- New Tradition Home's building science team facilitated its smooth transition by being open to any new improvements in energy efficiency building design.
- Because New Tradition is constantly learning, it had the background and the technical support to move ducts inside cost-effectively.
- New Tradition works with a willing and able HVAC subcontractor.
- Homes are two-story with 9-foot ceilings. (A 1-foot furred-down chase is not a significant intrusion on inside space.)
- New Tradition is already building quality homes using Building America best practices. This includes well sealed ducts and rim joists that ensure that ducts are actually within conditioned space.

Moving forward, New Tradition is examining all of its plans and redrafts to move ducts to inside spaces. As part of its systems approach, New Tradition is determining how fur-downs, dropped hallways, and other features that are needed to move ducts inside can be used to improve architectural elements within the home. For example, entering a bedroom with a 9-foot ceiling from a dropped hallway with an 8-foot ceiling gives the occupant a sense that the bedroom is larger than it is. The difference in ceiling height also serves to break up and define the spaces.

New Tradition will consider open-web trusses on future homes because open-web construction makes it easier to run ducts between floors and makes it easier for other trades to do their work, as explained above. This would require some changes in construction procedures, however.

Builders Roundup

Both Quadrant Homes and New Tradition Homes focused on getting most of the supply ducts in between floors. Their most important reasons for doing this were that the homes have two stories, and that the approach is cheap, doesn't require major redesigns, and is not a far leap from current practice. Quadrant, using a standardized approach, will place all of its air handlers in second story outside-space mechanical closets, regardless of plan type or furnace type. These techniques include moving conditioned space around the furnace in the garage and building an insulated closet around the furnace in the garage. It is important to note that if one uses this latter technique, it is critical to insulate the slab under the furnace. New Tradition dropped ceilings only where necessary, and only in hallways, closets, bathrooms, and small soffits in other living spaces. Quadrant, with its return in the attic and its open-web trusses, will be able to contain all ducts within the mechanical room and the attic, and between floors.

What About Other Techniques?

There are many options for moving ducts and HVAC equipment inside. Our research team was partial to the techniques discussed in this article. However, each home builder will have its own issues and practices, which may make the following options better suited to its particular situation (see Figure 2).

Cathedralized attic. A cathedralized attic is insulated at the roof deck, rather than above the top floor ceiling, or lid. When the cathedralized attic contains the ducts, it is not conditioned, nor is it vented to the outside, which creates a semiconditioned space. This technique is often used if the builder is trying to create a bonus room in the attic. However, it is difficult to insulate the roof truss to as high an R-value as the lid, because of the amount of space required to adequately insulate.

Often builders will use spray foam insulation in this case, but this is significantly more expensive than batt insulation. Other drawbacks include increased conditioned space and, most significantly, increased wall area coupled to the outside.

Plenum truss. The plenum truss approach utilizes adapted roof trusses that are designed to create semiconditioned space in the attic for ductwork. The plenum truss is a center section of the truss raised to provide a section for HVAC equipment and ducts. This design creates a plenum area in which supply ducts from the unconditioned attic are run. The interior of the plenum space is finished with an air barrier, isolating the ducts from the unconditioned space. Then the ceiling drywall and insulation are installed as appropriate. This method can be costly. (The photo above shows the air handler in the extended conditioned space of the garage.)

Building Forward

New Tradition Homes and Quadrant Homes are changing one of the important details in the design of their homes by moving the HVAC equipment inside. As the home market continues to tighten and Americans focus on energy efficiency, health, and greening their lifestyle, builders who are improving their homes today will be in a position to enjoy greater market success tomorrow. BIRA believes the techniques displayed by these two builders hold great promise for builders across the country. The methods discussed are cost-effective while offering benefits beyond energy savings, especially concerning the move to open-web trusses. Both builders have found systems that work for them—systems that exemplify how production builders can improve their homes in a cost-effective manner. We are proud to work with these builders, and we hope that America's production home builders take a page from their book to change the way their own homes operate, for the benefit of



The air handler is located inside the New Tradition home by extending the conditioned space into the garage.

themselves and their customers. It can be done.

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For more information:

Andrews, J. *Better Duct Systems for Home Heating and Cooling*. Building America, 2007. Can be downloaded from: www.eere.energy.gov.

Headrick, Roger. *Home Builders Guide to Ducts in Conditioned Space*. Park Ridge, IL. California Energy Commission, 2003. Can be downloaded from: www.energy.ca.gov/reports.

DOE's Building America program is a public/private partnership composed of six teams that provide energy solutions for production home builders. One of these teams is the Building Industry Research Alliance (BIRA). Led by ConSol Energy, BIRA is a group of building industry professionals dedicated to advancing home performance in communities whose goal is to produce as much energy as they consume. For more on Building America and BIRA, go to www.bira.ws.